

The Policy-Class Lattice: Which Restriction Destroyed Viability?

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Abstract

Which restriction destroyed viability — the codex, the reading, the instrument set, or the belief? The programme’s papers separate policy classes case by case; this paper, the D7 item of the generalization catalogue, organizes them as a lattice with a computable diagnostic. For policy classes on the two-patch working system, viability kernels are monotone under class inclusion, the chain institutional $\subseteq$ observation-based $\subseteq$ full-information holds setwise with exact counts $24 \subseteq 26 \subseteq 28$ of 36 safe pairs and is strict at both links, and the certainty-equivalence separation mechanism of the successor’s trap (corrected-class drift $\geq \frac{21}{100}$) is re-verified as the off-lattice separator. The diagnostic contribution is *distance to viability*: for the audited nonviable belief, the minimum sensor refinement that restores viability is exactly one split, the minimum action-set expansion is exactly one instrument (which restores viability under the unchanged reading, enlarging the viable family to all 36 pairs), and the minimum prior shrinking is exactly one state. Every number is machine-verified in exact arithmetic (6/6 check families).

Introduction and relation to the programme

The completed items distinguish perfect-information policies, observation-based policies, certainty-equivalence policies, and institutionally restricted policies as separate mechanisms. The source catalogue proposes the unified reading adopted here: a parameterized class Π with kernels $\text{Viab}_\Pi(\mathcal{V})$, monotone under inclusion, forming a hierarchy — and, more consequentially for governance, a diagnostic question: which restriction is responsible for the loss? The present paper instantiates that programme on the audited working system, computing the inclusions and the distances exactly. No completed item is modified; the stochastic selector’s probabilistic class and the successor’s five layers are cited where they bound the reading.

Policy classes and the chain

A policy class Π is a family of observation-based laws (measurable in the reading, acting per cell under the successor’s read-then-act semantics). Its kernel is the set of

beliefs from which some law of the class maintains the floors indefinitely.

Proposition 1 (Monotonicity). $\Pi_1 \subseteq \Pi_2 \implies \text{Viab}_{\Pi_1}(\mathcal{V}) \subseteq \text{Viab}_{\Pi_2}(\mathcal{V})$.

Proof. A witness law for B under Π_1 is a law of Π_2 . $\square$

Proposition 2 (The audited chain is strict). On the two-patch pair family: the institutional class (aggregate reading with a no-repeat codex) sustains 24 of 36 safe pairs; the per-cell aggregate class sustains 26; the full-information class sustains 28. Each inclusion is setwise and strict, with separating beliefs $\{(1, 2), (3, 1)\}$ at the lower link and $\{(1, 2), (2, 1)\}$ at the upper. On this family the certainty-equivalence class coincides with the per-cell class (laws act on the reading only); the successor’s trap corrected-class drift $\geq \frac{21}{100}$ on $s \in [1, 2]$ — is the mechanism that separates the two off this family.

Proof. The inclusions follow from monotonicity (each restricted class is contained in the next). The counts and the two separating beliefs are computed exactly by the verification script; the drift identity $g(s + \frac{1}{10}) - g(s) = \frac{s}{5} + \frac{1}{100} \geq \frac{21}{100}$ for $s \geq 1$ is exact. $\square$

Distance to viability

For a nonviable belief and a family of restorative moves, the *distance* is the minimum number of moves that restores viability.

Proposition 3 (Distances of the audited belief). For $B^* = \{(1, 2), (2, 1)\}$ under the aggregate reading (nonviable): (i) the minimum sensor refinement restoring viability is exactly one cell split — the unique one-split partition (the two singletons) is viable, and zero splits is the nonviable status quo; (ii) the minimum action-set expansion is exactly one instrument — adding $u = 3$ (protect both patches) restores viability of B^* under the *unchanged* aggregate reading and enlarges the aggregate viable family to all 36 pairs (setwise, strict); (iii) the minimum prior shrinking is exactly one state — either deletion yields a viable belief.

Proof. (i) A two-state belief admits exactly one partition into two nonempty cells; the script verifies the singleton partition viable and the zero-split (merged) verdict nonviable. (ii) With $u = 3$ the per-cell action sets of B^* ’s

single aggregate cell gain a jointly safe instrument; the recursion confirms viability, and the family count $26 \rightarrow 36$ is computed over all pairs. (iii) Both singletons are viable under the unchanged reading; the script verifies both and the nonviability of the whole. $\square$

Verified record (6/6 exact check families)

E1 the setwise chain with counts $24 \subseteq 26 \subseteq 28$; E2 strictness at both links plus the re-verified CE drift bound; E3 action-expansion monotonicity with counts $26 < 36$ and a strict instance; E4 the sensor-refinement distance = 1 certified by exhaustion of split counts; E5 the action-expansion distance = 1 with the restored belief; E6 the prior-shrinking distance = 1 with both deletions viable.

What is not claimed

The taxonomy is instantiated on one working system at one belief family (pairs): no general computational theory of distances is claimed, no continuous-state version, and no claim that one instrument of protection is a realistic policy lever — it is the minimal formal expansion. Institutional restrictions are modeled only as action codices; procedural and enforcement constraints belong to the programme’s decentralized- observation companion. The stochastic selector’s chance constraints are not revisited. The certainty-equivalence identification is stated for this family only, with the trap cited for the general separation.

Declarations

Funding. None. **Competing interests.** None. **Data availability.** No external data were used. **Code availability.** The verification script `policy_class_lattice_v1_verify.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it reproduces all six check families verbatim. **AI declaration.** GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review.